Reverse Physics

turning physics inside-out

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Assumptions of Physics

The foundations of physics has a methodology problem

Reverse Physics aims to adopt approaches that are more in line with foundations of mathematics, to break physics into a set of physical and mathematical conditions with well-understood logical relationships

Reverse Physics

"When the math is derived from the right physical assumptions, the physical assumptions can be derived from the math."

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Assumptions of Physics

Goals of Reverse Physics

- Framework/interpretation/formulation/theory neutral
  - Allow for each result to be independently formulated from the other
- Uncover “paradoxes” and develop standards of rigor appropriate for physics
- Clarify the logical dependencies of physical/mathematical conditions
- Identify core constitutive requirements of physical theories
  - The “big four” of Reverse Physics

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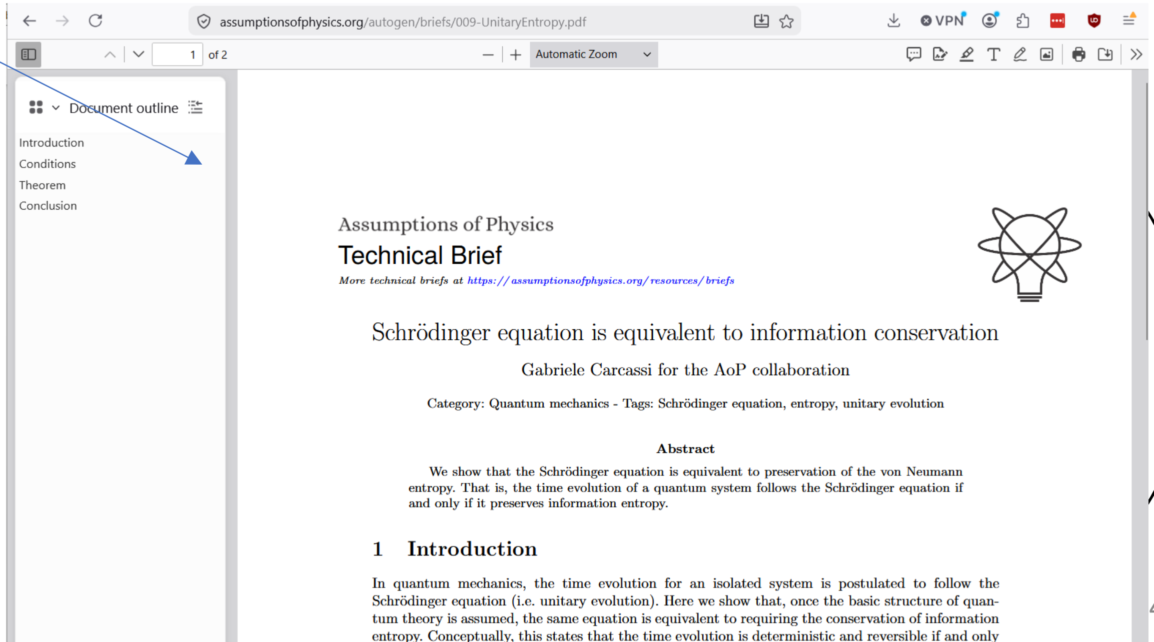
Assumptions of Physics

And now a series of random results to show how Reverse Physics works

Warning: no time to go through each in details

Wherever possible, QRCode to papers or technical briefs



The actual content of mathematical structures

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Assumptions of Physics

Hamiltonian mechanics 1 DOF $\Longleftrightarrow$ Det/Rev

$d_t\mu(U) = 0$ $\Longleftrightarrow$ $d_t \log \mu(U) = 0$ $\Longleftrightarrow$ $d_t \int -\rho \log \rho \, d\mu = 0$

Conservation of state count

Conservation of thermodynamic entropy

Conservation of information entropy

$d_t\omega_{ab} = 0$
symplectomorphism

Determinism and reversibility

$d_t\left|cov\left(\xi^a, \xi^b\right)\right| \approx 0$
(for peaked distributions)

$\Longleftrightarrow$ Conservation of uncertainty

$d_t q = \partial_p H$
$d_t p = -\partial_q H$
Hamilton's equations

All equivalent conditions!!!

Base theory:

State: $\xi^a = [\, q \; p \,] \in \mathbb{R}^2$

Count of states: $\mu(U) = \int_U dqdp = \int_U \omega$

Schrödinger equation $\Longleftrightarrow$ Det/Rev

$\psi(0) = \phi$ iff $\psi(t) = e^{\frac{H\Delta t}{\imath\hbar}}\phi$ $\Longleftrightarrow$ $d_t \mathrm{tr}(-\rho \log \rho) = 0$ $\Longleftrightarrow$ …

Bijective map of state and probability

Conservation of information entropy

…

$U(\Delta t)|\psi\rangle = e^{\frac{H\Delta t}{\imath\hbar}}|\psi\rangle$
unitary evolution

Determinism and reversibility

$\imath\hbar d_t|\psi\rangle = H|\psi\rangle$
Schrödinger equation

All equivalent conditions!!!

Base theory: State: $\psi \in P(\mathcal{H})$

Probability of ψ given ϕ: $p(\psi|\phi) = |\langle\psi|\phi\rangle|^2$

Reversing the principle of stationary action

DR

$$\nabla \cdot \vec{S} = 0$$

No state is "lost" or
"created" as time evolves

$[p, 0, -H(q,p)]$

$$\vec{S} = -\nabla \times \vec{\theta}$$

(Minus sign to recover Ham eq)

KE

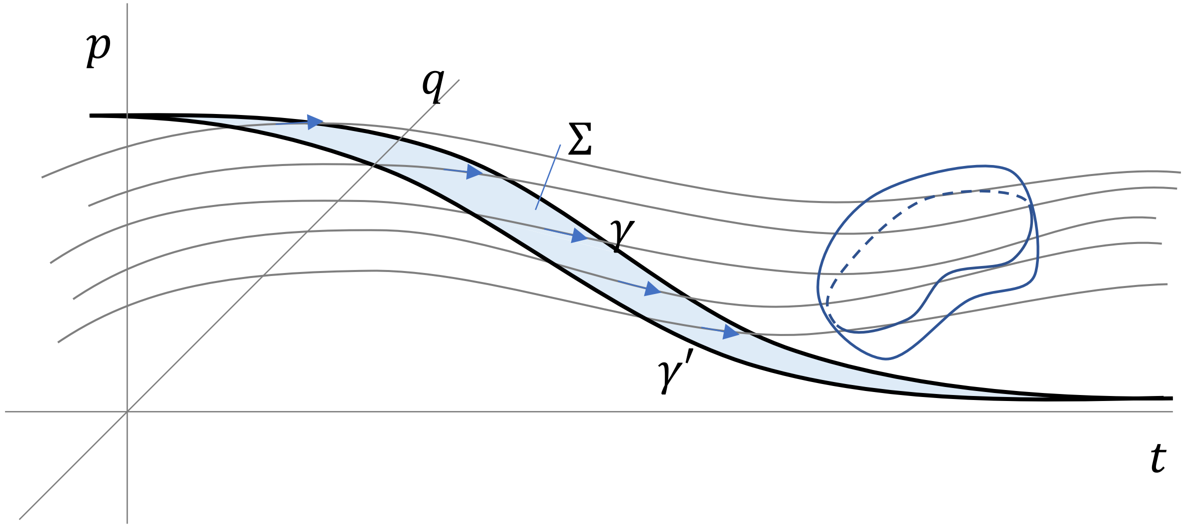
$p\frac{dq}{dt} + 0\frac{dp}{dt} - H\frac{dt}{dt}$

$$\mathcal{A}[\gamma] = \int_\gamma L dt = \int_\gamma \vec{\theta} \cdot d\vec{\gamma}$$

Sci Rep **13**, 12138 (2023)

unphysical

The action is the line integral of the vector potential of the flow of states



Variation of the action

Gauge independent,
physical!

$$\delta\mathcal{A}[\gamma] = \oint_{\partial\Sigma} \vec{\theta} \cdot d\vec{\gamma}$$

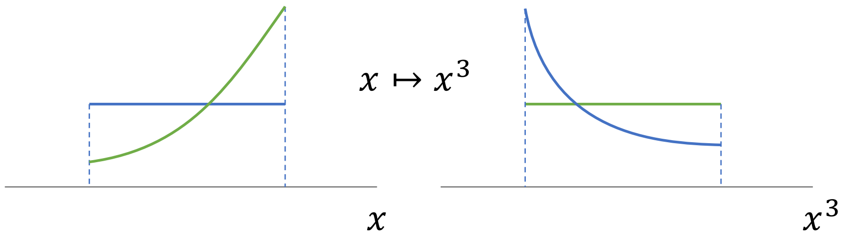
$$= -\iint_\Sigma \vec{S} \cdot d\vec{\Sigma}$$

Variation of the action measures the flow of states (physical).
Variation = 0 ⇒ flow of states tangent to the path.

Reversing phase-space

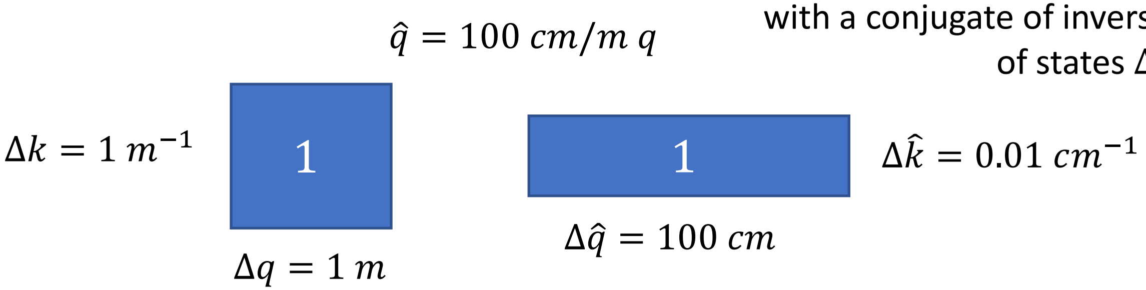
Hamiltonian Privilege, *Erkenn* (2023)

Stud Hist Phil **71**, 082020, 60-71 (2020)



Density, entropy, uniform distributions NOT in general coordinate invariant

Each unit variable (i.e. coordinate) paired with a conjugate of inverse units: number of states $\Delta q \Delta k$ is invariant



Phase space (symplectic) structure is the only one that supports coordinate invariant density, entropy, state count

Phase space volumes are invariant under equal-time changes of coordinates q^i	(PSV-VOL)
The Jacobian for the transformation induced by equal-time changes of coordinates q^i is unitary	(PSV-JAC)
Densities over phase space are invariant under equal-time changes of coordinates q^i	(PSV-DEN)
Thermodynamic entropy is invariant under equal-time changes of coordinates q^i	(PSV-THER)
Information entropy is invariant under equal-time changes of coordinates q^i	(PSV-INFO)
Uncertainty of peaked distributions is invariant under equal-time changes of coordinates q^i	(PSV-UNC)

The system allows statistically independent distributions over each DOF under any choice of coordinates q^i	(PSI-DEN)
The system allows informationally independent distributions over each DOF under any choice of coordinates q^i	(PSI-INFO)
The system allows peaked distributions where the uncertainty is the product of the uncertainty on each DOF under any choice of coordinates q^i	(PSI-UNC)

Invariance of entropy/state count + DOF independence $\Longleftrightarrow$ structure of phase space (i.e. conjugate pairs)

Geometry is entropy

Classical mechanics

$$S(\rho_U) = \log \mu(U) = \log \int_U \omega$$

Symplectic form determines and is determined by the entropy of all uniform joint and marginal (DOF) distributions

$$\partial_\alpha g_{\beta\gamma} - \partial_\beta g_{\alpha\gamma}$$

$$\omega_{ab} = \begin{bmatrix} -mG_{\alpha\beta\gamma}u^\gamma - qF^{\alpha\beta} & m g_{\alpha\beta} \\ -m g_{\alpha\beta} & 0 \end{bmatrix}$$

Metric tensor appears as the position-velocity part of the symplectic form

Quantum mechanics

$$S\left(\frac{1}{2}\rho_\psi + \frac{1}{2}\rho_\phi\right) = I\left(\frac{1+|\langle\psi|\phi\rangle|}{2}, \frac{1-|\langle\psi|\phi\rangle|}{2}\right)$$

$$= -\frac{1+|\langle\psi|\phi\rangle|}{2}\log\frac{1+|\langle\psi|\phi\rangle|}{2} - \frac{1-|\langle\psi|\phi\rangle|}{2}\log\frac{1-|\langle\psi|\phi\rangle|}{2}$$

Born rule determines and is determined by the entropy of all equal mixtures of pairs of pure states

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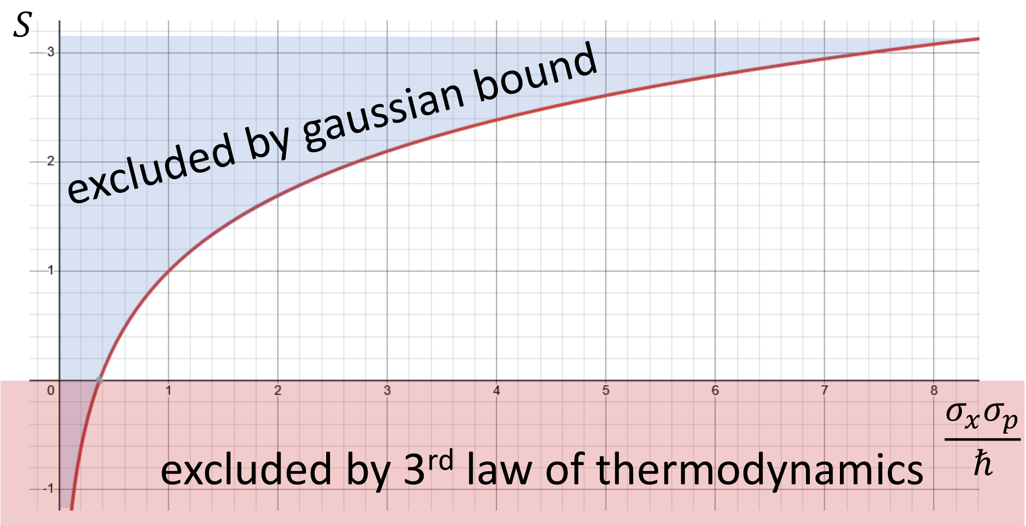
Assumptions of Physics

Classical uncertainty principle

$$S(\rho) = -\int \rho \log h\rho \, dq dp$$

Fixes units

Uniform distribution over volume h has zero entropy



$$S(\rho) \leq \log 2\pi e \frac{\sigma_q \sigma_p}{h}$$

Gaussian maximizes entropy for a given uncertainty

$$\sigma_q \sigma_p \geq \frac{h}{2\pi e} e^{S(\rho)} = \frac{\hbar}{e} e^{S(\rho)}$$

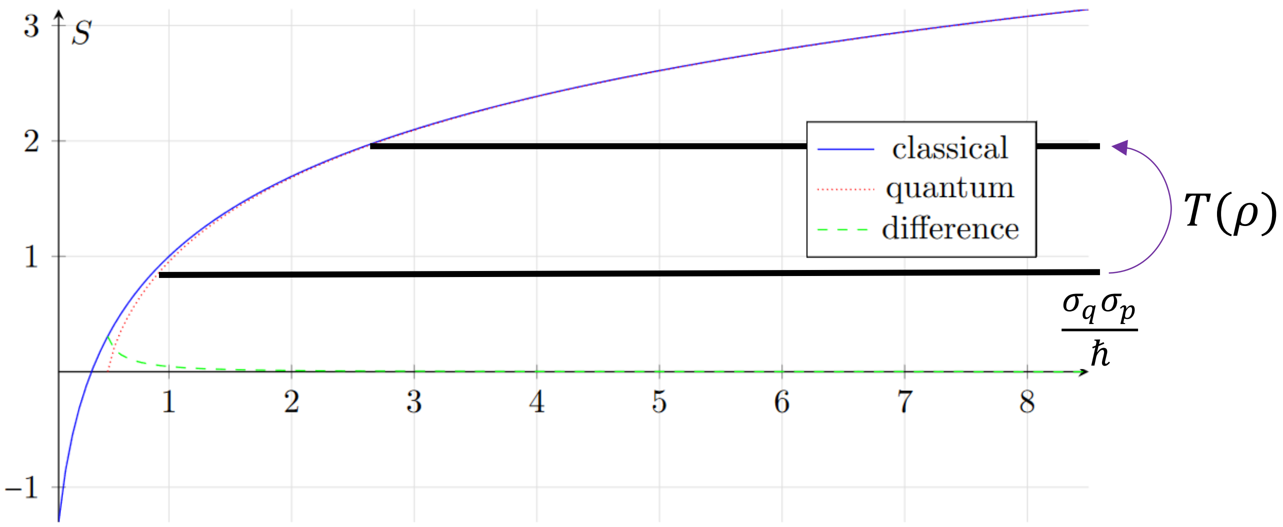
Entropy puts a lower bound on the uncertainty

$$S \geq 0 \quad \Rightarrow \quad \sigma_q \sigma_p \geq \frac{\hbar}{e}$$

Classical mechanics as high entropy limit

Physical limit: $S \gg 0$

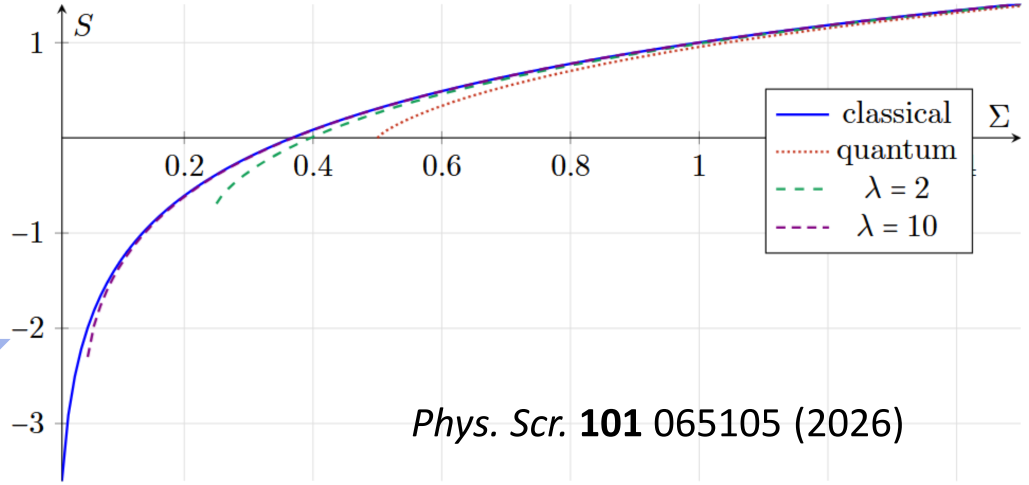
like $\frac{v}{c} \ll 1$



$[X, P] = \imath\hbar \qquad [T(X), T(P)] = \lambda \imath \hbar$

Mathematical limit: $\hbar \to 0$

like $c \to \infty$



Phys. Scr. **101** 065105 (2026)

$[T(\hat{X}), T(\hat{P})] = \imath\hbar$

$[\hat{X}, \hat{P}] = \frac{\imath\hbar}{\lambda}$

Physical limit: increase the entropy of mixed states

Mathematical limit: decrease the entropy of pure states

Non-linear representation in QM

This map introduces a total phase based on all coefficients

const. $\theta \in (0,2\pi)$

$$m(c_+|z^+\rangle + c_-|z^-\rangle) = e^{\iota\theta\frac{|c_+|^2}{|c_+|^2+|c_-|^2}}(c_+|z^+\rangle + c_-|z^-\rangle)$$

Non-linear map…

… that preserves the physics
(i.e. Born rule)

$$\frac{\langle m(\psi)|m(\phi)\rangle\langle m(\phi)|m(\psi)\rangle}{\langle m(\psi)|m(\psi)\rangle\langle m(\phi)|m(\phi)\rangle} = \frac{\langle\psi|\phi\rangle\langle\phi|\psi\rangle}{\langle\psi|\psi\rangle\langle\phi|\phi\rangle}$$

Linearity is a mathematical convenience,
not a physical necessity

Superposition is multi-decomposition

Property of the **vector space representation**

Superposition: $|\phi\rangle = c_1|\psi_1\rangle + c_2|\psi_2\rangle$

Impossible in classical probability
(i.e. Choquet simplex)

$$\Updownarrow$$

Multi-decomposition: $\rho = p_\phi \rho_\phi + p_{\hat{\phi}} \rho_{\hat{\phi}} = p_1 \rho_{\psi_1} + p_2 \rho_{\psi_2}$

Property of **distinguishability of preparations**

Linear transformations

$$\Updownarrow$$

Output of mixtures = mixture of outputs

Indistinguishability of different mixtures

$$\Updownarrow$$

No physical process can depend on them

Paradoxes and standards of rigor

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Assumptions of Physics

Poor dimensional checking in statistical mechanics

Definition of Gibbs entropy

$$-\int_X \rho \log \rho \, dq^i \, dp_j \quad \Rightarrow -\int_X \rho \log h^n \rho \, dq^i \, dp_j$$

Density is not a pure number

When deriving Sackur–Tetrode equation

$$\Omega_N = \frac{1}{N!}\frac{V^N}{h^{3N}}\frac{2\pi^{3N/2}}{(\frac{3N}{2}-1)!}(\sqrt{2mU})^{3N-1} \approx \frac{1}{N!}\frac{V^N}{h^{3N}}\frac{\pi^{3N/2}}{(3N/2)!}(\sqrt{2mU})^{3N}.$$

You can't "approximate physical dimensions"!!!

$$length^N \neq length^{N-1}$$

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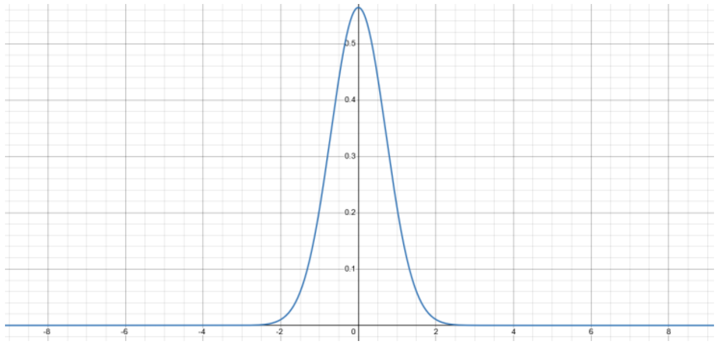
Assumptions of Physics

Inconsistent expectations in Hilbert spaces

Quantum Stud.: Math. Found. **12**, 13 (2025)

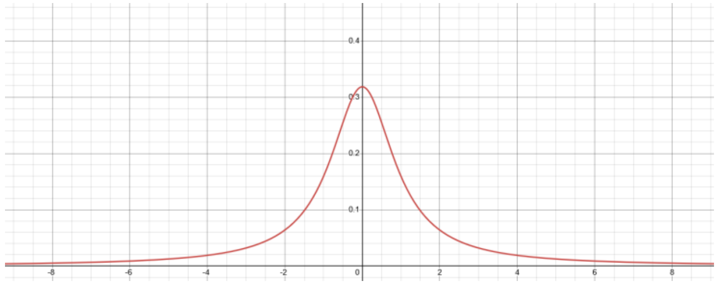
$$\psi(x) = \sqrt{\frac{e^{-x^2}}{\sqrt{\pi}}} \qquad \int |\psi|^2 dx = 1$$

$$\rho_\psi(x) = \frac{e^{-x^2}}{\sqrt{\pi}} \qquad \langle X^2 \rangle_\psi = \frac{1}{2}$$



$$\phi(y) = \sqrt{\frac{1}{\pi(y^2+1)}} \qquad \int |\phi|^2 dy = 1$$

$$\rho_\phi(y) = \frac{1}{\pi(y^2+1)} \qquad \langle Y^2 \rangle_\phi \to \infty$$



Different observers see finite/infinite expectation

$$y = \tan\left(\frac{\pi}{2}\operatorname{erf}(x)\right)$$

$$\psi(y) = \psi(x)\sqrt{\frac{dx}{dy}}$$

Expectation can have finite/infinite oscillations

$$x(x_0, t) = x_0 \cos^2\frac{\pi t}{2} + \tan\left(\frac{\pi}{2}\operatorname{erf}(x_0)\right)\sin^2\frac{\pi t}{2}$$

Every continuous linear operator defined on the whole Hilbert space is bounded ⇒ position/momentum/energy/number of particles are not defined on the whole Hilbert space!!!

Inconsistent convergence in Hilbert spaces

Consider the sequence of states given by

$$|\psi_i\rangle = \sqrt{\frac{i-1}{i}}|0\rangle + \sqrt{\frac{1}{i}}|i\rangle$$

$$\langle\psi_i|N|\psi_i\rangle = \frac{i-1}{i}\langle 0|N|0\rangle + \frac{1}{i}\langle i|N|i\rangle \to 1$$

$$|\psi_i\rangle \to |0\rangle$$

The state converges to the ground state
while the energy level converges to the first level

Inconsistent units for quantum observables

Suppose X is an observable

Expectation $\langle\psi|X|\psi\rangle$ should have units of X

But then $X|\psi\rangle$ cannot have the same units of $|\psi\rangle$

But then $X|\psi\rangle + |\psi\rangle$ makes no physical sense

Therefore, X cannot be a map within the same space

Bonus: what are the units for $\psi(x)$?

Then why in some relativistic settings do they use the
Fourier transform for a scalar function?

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Assumptions of Physics

Tangent vectors in differential geometry

Defined as derivations

$$v: C^{\infty}(X) \to \mathbb{R}$$

$$v = \sum_i v^i \partial_i$$

component basis

In polar coordinates

$$\partial_r + \partial_\theta = ???$$

$[\mathrm{m}]$ $[\mathrm{rad}]$

In phase space

$$\partial_q + \partial_p = ???$$

$[\mathrm{m}]$ $[\mathrm{kg\ m\ s^{-1}}]$

Mathematically precise $\not\Rightarrow$ physically precise

Doesn't work with units

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Assumptions of Physics

The Born rule is equivalent to the von Neumann entropy

Condition EQST (Ensembles and quantum states). *The space of ensembles $E(\mathcal{H})$ of a quantum system is represented by the convex space of all density operators (i.e positive semi-definite trace one self-adjoint operators) over a suitable separable inner product space $\mathcal{H}$. The state space is the set of extreme points, which coincides with the projective space $P(\mathcal{H})$. The ensemble space is equipped with a function $S : E(\mathcal{H}) \to \mathbb{R}$ where $S(\rho)$ returns the entropy for ensemble ρ. The ensemble space is also equipped with a function $p(\cdot|\cdot) : P(\mathcal{H}) \times E(\mathcal{H}) \to [0,1]$, where $p(\psi|\rho)$ returns the probability of finding ψ having prepared ρ. This function is affine in the second argument (i.e. $p(\phi|\sum p_i\psi_i) = \sum_i p_i p(\phi|\psi_i)$) and returns one if and only if the initial and final states are the same (i.e. $p(\phi|\psi) = 1 \iff \phi = \psi$ for all $\psi, \phi \in P(\mathcal{H})$).*

Condition BR-BORN (Born rule). *The conditional probability p is given by the Born rule: $p(\psi|\phi) = \frac{\langle\phi|\psi\rangle\langle\psi|\phi\rangle}{\langle\psi|\psi\rangle\langle\phi|\phi\rangle}$.*

Condition BR-ENT (Von Neumann entropy). *The entropy function S is given by the Shannon entropy: $S(\rho) = -\operatorname{tr}(\rho \log \rho)$.*

Condition [B]MUTEX (Mutual exclusivity conditions). *Let $\psi_i \in P(\mathcal{H})$ be a countable sequence of states. Then $p(\psi_i|\psi_j) = 0$ for all $i \neq j$ if and only if $S(\sum_i p_i\psi_i) = -\sum_i p_i \log p_i$ for all p_i such that $\sum_i p_i = 1$.*

Theorem 9. *Over EQST, if two conditions among BR-BORN, BR-ENT and [B]MUTEX are assumed, the third is implied.*

We CAN analyze logical dependencies between conditions that are both mathematically precise and physically meaningful

Logical dependencies and base theories

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Assumptions of Physics

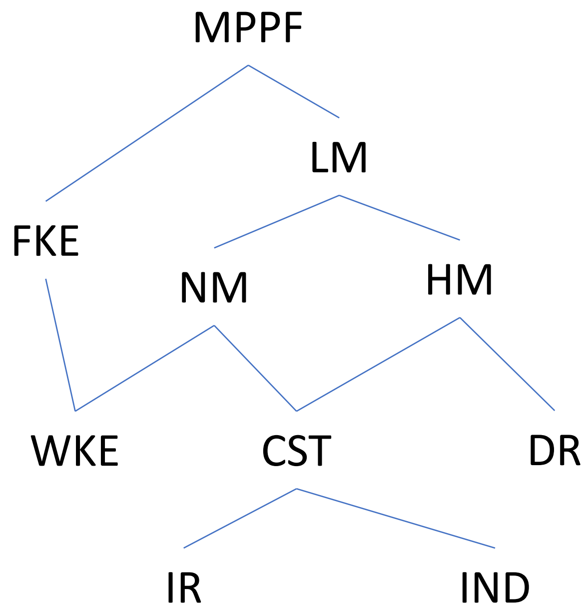
Classical conditions and their relationships

Core assumptions

IR: Infinitesimal Reducibility
IND: INDependent DOF
W/FKE: Weak/Full Kinematic Equivalence
DR: Determinism+Reversibility

Physical theories

CST: Classical STates
NM: Newtonian Mechanics
HM: Hamiltonian Mechanics
LM: Lagrangian Mechanics
MPPF: Massive Particles under Potential Forces



Each assumption is an equivalence class of multiple conditions

There is a linear relationship between conjugate momentum and velocity (FKE-LIN)

At each position, there exists a local inertial frame (FKE-INER)

The position fully defines the units of all state variables, and therefore an invertible transformation between momentum and velocity (FKE-UNIT)

Density expressed in velocity at the same position is proportional to the density over states (FKE-PROP)

Uniform distributions along momentum correspond to uniform distributions along velocity (FKE-UNIF)

The system under study is a massive particle under scalar and vector potential forces (FKE-POT)

The Jacobian of the transformation between state variables and kinematic variables is a non-singular function of position only. (FKE-NSIN)

Densities over phase space can be expressed in terms of position and velocity by rescaling the value at each point: $\rho(x^i, v^j)|J(x^i)| = \rho(q^i, p_j)$. (FKE-DEN)

Areas and volumes in phase space can be expressed in kinematic variables, and the transformation depends on position only: $dx^1 \cdots dx^n\, dv^1 \cdots dv^n = |J(x^i)|\, dq^1 \cdots dq^n\, dp_1 \cdots dp_n$. (FKE-VOL)

The symplectic form ω_{ab} can be expressed in kinematic variables, and its components are a linear function of velocity. (FKE-SYMP)

Quantum pure states are not enough

Condition QP-RAY (Quantum states are rays). *The state space of the system is represented by the projective space given by the rays of a complex vector space* $\mathcal{H}$. *That is, a state* $\psi \in P(\mathcal{H})$ *is a ray* $\psi = \{c|\psi\rangle \mid c \in \mathbb{C}\}$ *of a complex projective space.*

Rays do not need the inner product

Condition CST (Classical state space). *The state space is represented by a symplectic manifold* (X, ω), *typically charted by conjugate quantities (i.e. local Darboux coordinates)* $(q_i, p_i) \in \mathbb{R}^{2n}$, *where* n *is the number of independent degrees of freedom (DOF). The count of configurations for each DOF over a region* Σ *is given* $\int_\Sigma dp^i dq_i = \int_\Sigma \omega$ *and the count of states is given by the Liouville measure* $\mu(U) = \int_U dq^1 \cdots dq^n dp_1 \cdots dp_n = \int_U \omega^n$.

The ensemble space is represented by th set of probability distributions $\rho(q^i, p_i)$ *(i.e. probability measures that are absolutely continuous with respect to the Liouville measure).*

The conditional probability over states is given by $p(\xi_1|\xi_2) = 0 = p(\xi_2|\xi_1)$ *for all* $\xi_1, \xi_2 \in X$, *meaning that all states represent mutually exclusive events.*

Theorem 1. *Conditions QP-RAY and CST are consistent. In fact, they are consistent if and only if* $1 \leq \dim(\mathcal{H}) < \infty$.

You CAN endow quantum pure states with classical probability distributions and Hamiltonian mechanics

E.g. Classical Larmor precession

Quantum ensembles rule out classical mechanics

Condition QE-DENS (Quantum ensembles are density operators). *The ensemble space of the system is represented by the set of density operators acting on a complex inner product space $\mathcal{H}$. That is, an ensemble $\rho \in D(\mathcal{H})$ is a positive semi-definite trace one self-adjoint operator $\rho : \mathcal{H} \to \mathcal{H}$.*

Trace requires the inner product!

Condition BR-BORN (Conditional expectation from Born rule). *The conditional probability $p : P(\mathcal{H}) \times P(\mathcal{H}) \to [0,1]$ is given by the Born rule. That is, $p(\psi|\phi) = \frac{\langle\phi|\psi\rangle\langle\psi|\phi\rangle}{\langle\psi|\psi\rangle\langle\phi|\phi\rangle}$.*

Theorem 3. *Conditions QE-DENS and CST are inconsistent.*

Classical and quantum ensembles are incompatible

Since BR-BORN is independent from QE-DENS,
QE-DENS is responsible for the incompatibility

Base theory for all of physics

Collection of conditions that MUST be true in all physical theories

Condition EQST (Ensembles and quantum states). *The space of ensembles $E(\mathcal{H})$ of a quantum system is represented by the convex space of all density operators (i.e positive semi-definite trace one self-adjoint operators) over a suitable separable inner product space $\mathcal{H}$. The state space is the set of extreme points, which coincides with the projective space $P(\mathcal{H})$. The ensemble space is equipped with a function $S : E(\mathcal{H}) \to \mathbb{R}$ where $S(\rho)$ returns the entropy for ensemble ρ. The ensemble space is also equipped with a function $p(\cdot|\cdot) : P(\mathcal{H}) \times E(\mathcal{H}) \to [0,1]$, where $p(\psi|\rho)$ returns the probability of finding ψ having prepared ρ. This function is affine in the second argument (i.e. $p(\phi|\sum p_i\psi_i) = \sum_i p_i p(\phi|\psi_i)$) and returns one if and only if the initial and final states are the same (i.e. $p(\phi|\psi) = 1 \iff \phi = \psi$ for all $\psi, \phi \in P(\mathcal{H})$).*

Condition [B]MUTEX (Mutual exclusivity conditions). *Let $\psi_i \in P(\mathcal{H})$ be a countable sequence of states. Then $p(\psi_i|\psi_j) = 0$ for all $i \neq j$ if and only if $S(\sum_i p_i\psi_i) = -\sum_i p_i \log p_i$ for all p_i such that $\sum_i p_i = 1$.*

Part of the base theory

Base theory ≠ Theory of everything

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Assumptions of Physics

Foundational pillars

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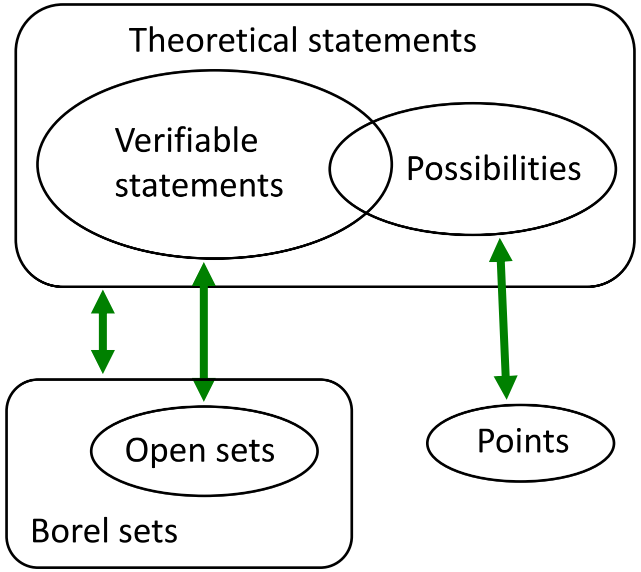
Assumptions of Physics

Experimental verifiability

Science is evidence based ⇒ Statements are associated to experimental tests

Verifiable statement	Test Result
T	SUCCESS (in finite time)
F	UNDEFINED
	FAILURE (in finite time)



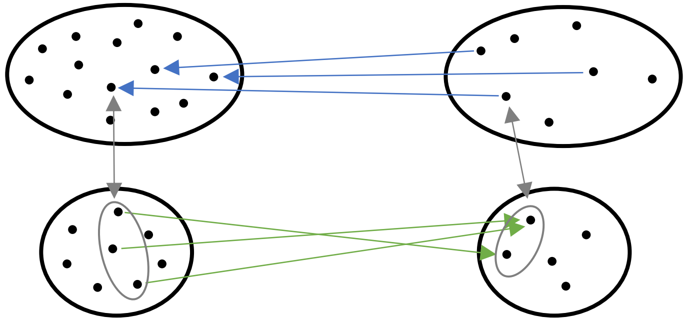
s	Test
T	S
	U
F	U
	F

$int(A)$ corresponds to the verifiable part of a statement

∂A corresponds to the undecidable part of a statement

$ext(A)$ corresponds to the falsifiable part of a statement

Inference relationship $r: \mathcal{D}_Y \to \mathcal{D}_X$ such that $r(s) \equiv s$



Causal relationship $f: X \to Y$ such that $x \preccurlyeq f(x)$

Experimental verifiability ⇒ topology and σ-algebra

Experimental relationships must be topologically continuous

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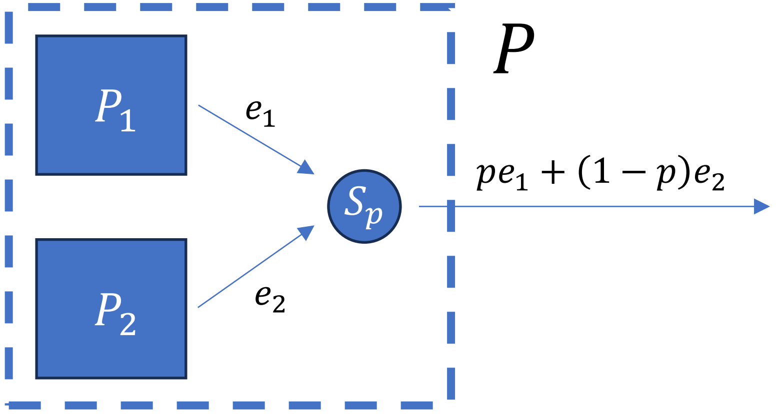
Assumptions of Physics

Reproducibility and ensembles

Scientific laws describe reproducible relationships $\Rightarrow$ Ensembles



Ensembles are outputs of repeatable processes

Statistical mixing $\Rightarrow$ Linear structures in physics

Note: mixing coefficients are NOT probabilities

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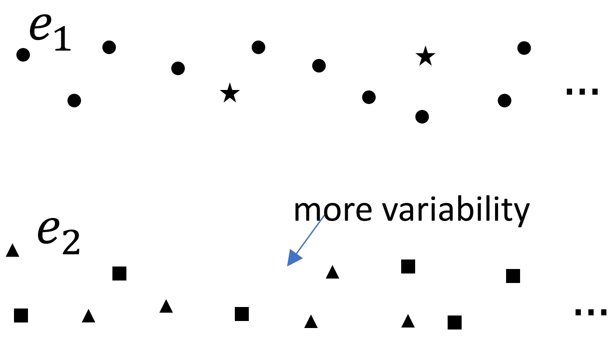
Assumptions of Physics

Ensemble variability

Variability of the elements within an ensemble ⇒ Entropy



Mutually exclusive ⇔ orthogonality

⇕

Maximal variability increase during mixing

Variability ⇒ Probabilities, inner product and geometry

Probability ⇔ Mixtures of mutually exclusive ensembles

Classical mechanics/probability ⇔ All pure states are mutually exclusive

Pure states ⇔ Best preparations ⇎ No variability

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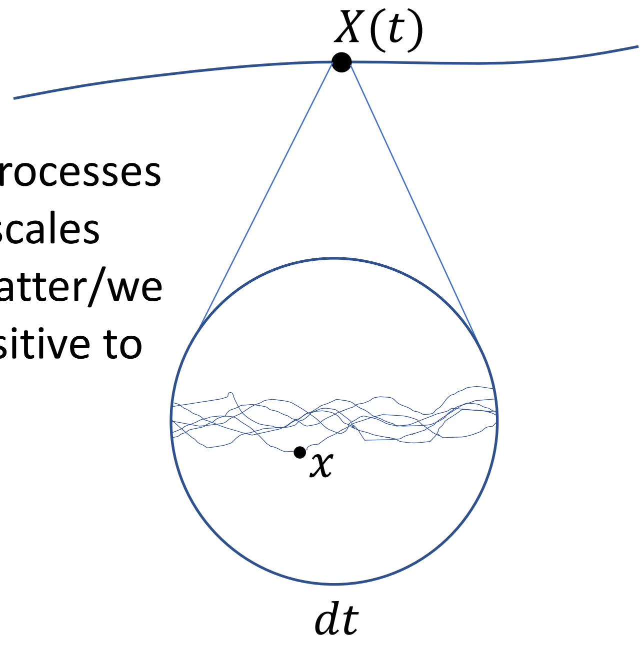
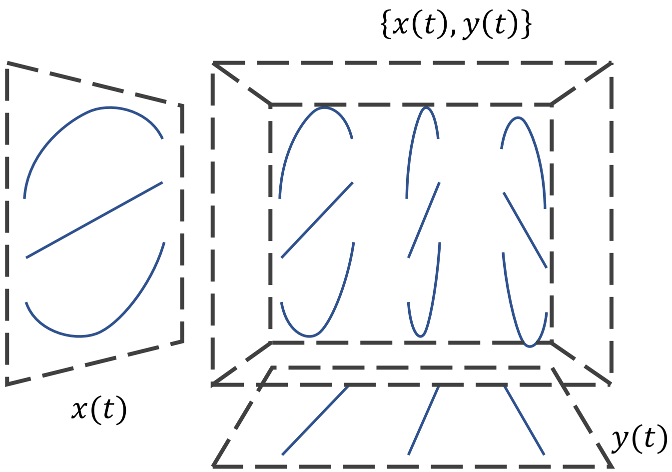
Assumptions of Physics

Equilibria

States must exist for at least a short time ⇒ Equilibria

Of internal processes
at faster scales
that don’t matter/we
are not sensitive to





Of system/environment
interactions
that don’t matter/we
are not sensitive to

Least developed part

This requirement allows us to properly talk about different physical objects

Equilibria ⇒ Symmetries, differentiability, Lie algebras

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Assumptions of Physics

The foundations of physics has a methodology problem

Reverse Physics gives us a powerful methodology, more in line with foundational approaches of other disciplines, that gives us a lot more and more precise insights and opens new avenues of research

Reverse Physics

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Assumptions of Physics

Working slide

- We can better understand the physical

https://assumptionsofphysics.org/

Assumptions of Physics